Shengqi Qiu

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Education

University of Wisconsin–Madison Madison, WI
B.S. in Mathematics and Computer Science *Sep 2024 – Dec 2026*

- **GPA: 3.638/4.000**

Skills

- **Languages:** English (fluent), Mandarin (native).
- **Programming Languages:** Java, Python, C++/C, CSS, JavaScript, R, SQL, Bash, Assembly(LC-3)
- **Mathematics:** Probability, Stochastic Processes, Topology, Abstract Algebra, Commutative Algebra, Linear Algebra, Mathematical Analysis, Numerical Analysis.

Research Experience

- **Undergraduate Researcher with Professor Frederic Sala** Feb 2026 – Present
  - **Program-As-a-Judge Automated Model Assessment (Pajama)** Feb 2026 – Jul 2026
    - Developed and implemented **Pajama**, a **hybrid program–LLM evaluation system** that distills LLM judging logic into executable Python programs, aggregates calibrated verdicts via weak supervision, and selectively routes uncertain cases to LLM judges.
    - Developed confidence-aware routing leveraging Snorkel Label Model posteriors to selectively escalate low-confidence cases to LLM judges, advancing the accuracy–throughput Pareto frontier; achieved **+5.0% accuracy at 2.9× throughput** over LLM-only evaluation.
    - Matched the accuracy of a **13B** LLM judge at **47.25× higher throughput** using programmatic judges; distilled reward models from program-generated preference labels that outperformed GPT-4-labeled baselines on RewardBench at **45–50×** lower API labeling cost.
  - **The Effect of Layer Ordering in Hybrid Architectures** Jun 2026 – Sep 2026
    - Proposed two novel tasks, **Segmented Retrieval** and **Exact Copy**, specifically designed to demonstrate the advantages of Transformer-first (TS) over SSM-first (ST) hybrid architectures under resource constraints.
    - Proved **recurrent-state memory lower bounds** for SSM-first hybrids using prefix–suffix distinguishability: $\Omega(\lambda \log |\mathcal{N}|)$ bits for Segmented Retrieval and $\Omega(\ell \log |\mathcal{V}|)$ for Exact Copy under bounded attention windows; constructed efficient TS architectures requiring only logarithmic recurrent state.
    - Extended the lower bounds to convolutional SSMs, including standard Mamba, deriving a **tight accumulated-lookback threshold** that characterizes when convolution can bypass recurrent-state bottlenecks; and contributed to verifying the theoretical proofs throughout the paper.
    - Conducted fixed-inference-memory experiments across all four two-layer orderings (TT, TS, ST, SS), TS achieved **100% accuracy** on Segmented Retrieval and **99%** on Exact Copy, compared with **below 50%** and **62%**, respectively, for the best-performing competing two-layer architecture on each task.

- **Undergraduate Researcher with Professor Benedek Valkó** Jan 2026 – Jul 2026
  Madison Experimental Mathematics Lab (MXM)
  - **The Pair Correlation Function of the Sine_6 Process**
    - Independently discovered an **operator factorization** reducing the third-order ODE arising from the Qu–Valkó framework for the Sine_6 process to solvable lower-order equations.
    - Derived the **first explicit single-variable Bessel-function representation** of the Sine_6 pair correlation function, providing a one-dimensional integral alternative to Forrester's six-dimensional Selberg integral.
    - Verified the large-separation asymptotics of the new representation against Forrester's classical results, including higher-order oscillatory corrections.

- **Undergraduate Researcher with Professor Grigorios Chrysos** Oct 2025 – Apr 2026
  - **Robustness Verification for Transformer Models**
    - Derived a **spectral-norm Jacobian bound** for dot-product self-attention as a step toward extending certified robustness methods beyond convolutional text classifiers.

Publications and Preprints

(denotes equal contribution)*

1. T.-H. Huang*, **S. Qiu*** and F. Sala. **"Codifying the Judge: Scalable Evaluation via Program Distillation."** *Under Review*, 2026. URL: https://arxiv.org/abs/2607.22561
2. J. Cooper*, M. Ma*, J. Mo*, **S. Qiu***, N. Bastian, I. Diakonikolas, and F. Sala. **"The Effect of Layer Ordering in Hybrid Architectures."** *Under Review*, 2026. URL: https://arxiv.org/abs/2610.18104
3. **S. Qiu**, Y. Qu, L. Yuan, B. Valkó, and S. Venancio. **"The Pair Correlation Function of the Sine_6 Process."** *Under Review*, 2026. URL: https://arxiv.org/abs/2607.26223
4. S. Gu, **S. Qiu**, H. Zhou, and X. Luo. **"Retrieval-Guided Twin Fusion with Similarity-Aware Contrast for Molecule-Text Alignment."** *Under Review*, 2026. URL: https://arxiv.org/abs/2608.16005

Relevant Coursework

- **Honors Courses in Math:** MATH 541 *Modern Algebra I*, MATH 542 *Modern Algebra II*, MATH 521 *Analysis I*, MATH 522 *Analysis II*, MATH 551 *Elementary Topology*, MATH/CS 514 *Numerical Analysis*.
- **Other Advanced Mathematics:** MATH 632 *Stochastic Processes*.
- **Computer Science:** CS 514 *Numerical Analysis*, CS 540 *Intro to AI*, CS 577 *Intro to Algorithms*, CS 639 *Intro to Foundation Models*, CS 544 *Intro to Big Data Systems*, CS 564 *Databases*.

Awards

Dean's List for Fall 2025